Task

Read the statements below.

For each statement decide which state of matter has made it.

Hint: Some statements could be made by more than one state of matter.

1. I have a fixed volume.
2. If I am cooled to my boiling point I will condense.
3. I can take the shape of whichever container I am poured into.
4. My particles move freely and randomly.
5. If I am heated to my boiling point I will evaporate.
6. My particles vibrate around a fixed position.
7. My particles can spread out to fill whatever space they are in.
8. If you heat me, my particles will gain energy.
9. I have a fixed shape.
10. I have a fixed volume, but not a fixed shape.

Challenge task

Write a ‘who am I’ sentence of your own.

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Teaching notes and answers

This is a KS3 activity for states of matter. It can be used as an AfL activity at various points in teaching the topic. It is easily self or peer assessed.

It can also be used as a starter/recap activity for weaker KS4 groups and used to identify and challenge misconceptions.

1. Solid and liquid
2. Gas
3. Liquid and gas
4. Gas
5. Liquid
6. Solid
7. Gas
8. Solid, liquid and gas
9. Solid
10. Liquid